1.15 Volume of Revolution

Worked Solutions

1 a)

$y^2 = (x+1)^2 = x^2 + 2x + 1$

1 b)

$y^2 = (3x-1)^2 = 9x^2 - 6x + 1$

1 c)

$y^2 = \left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2}$

1 d)

$y^2 = (x^2+5)^2 = x^4 + 10x^2 + 25$

1 e)

$y^2 = (2\sqrt{x}+3)^2 = 4x + 12\sqrt{x} + 9$

1 f)

$y^2 = \left(\frac{1}{\sqrt{x^3}} + 1\right)^2 = \frac{1}{x^3} + \frac{2}{\sqrt{x^3}} + 1$

2 a)

$$\begin{aligned}
&\pi \int_1^5 y^2 \,\mathrm{d}x = \pi \int_1^5 (x-1)^2 \,\mathrm{d}x = \pi \int_1^5 (x^2 - 2x + 1) \,\mathrm{d}x \\
&= \pi \left[\frac{1}{3}x^3 - x^2 + x\right]_1^5 \\
&= \pi \left[\left(\frac{5^3}{3} - 5^2 + 5\right) - \left(\frac{1^3}{3} - 1^2 + 1\right)\right] \\
&= \frac{64}{3}\pi \text{ units}^3
\end{aligned}$$

2 b)

$$\begin{aligned}
&\pi \int_0^2 y^2 \,\mathrm{d}x = \pi \int_0^2 (x^2+2)^2 \,\mathrm{d}x = \pi \int_0^2 (x^4 + 4x^2 + 4) \,\mathrm{d}x \\
&= \pi \left[\frac{1}{5}x^5 + \frac{4}{3}x^3 + 4x\right]_0^2 \\
&= \pi \left[\left(\frac{2^5}{5} + \frac{4 \times 2^3}{3} + 8\right) - (0)\right] \\
&= \frac{376}{15}\pi \text{ units}^3
\end{aligned}$$

2 c)

$$\pi \int_1^3 y^2 \, \mathrm{d}x = \pi \int_1^3 (x^{-\frac{3}{2}})^2 \, \mathrm{d}x = \pi \int_1^3 x^{-3} \, \mathrm{d}x$$
$$= \pi \left[-\frac{1}{2} x^{-2} \right]_1^3$$
$$= \pi \left[\left(-\frac{1}{2} \times 3^{-2} \right) - \left(-\frac{1}{2} \times 1^{-2} \right) \right]$$
$$= \frac{4}{9} \pi \text{ units}^3$$

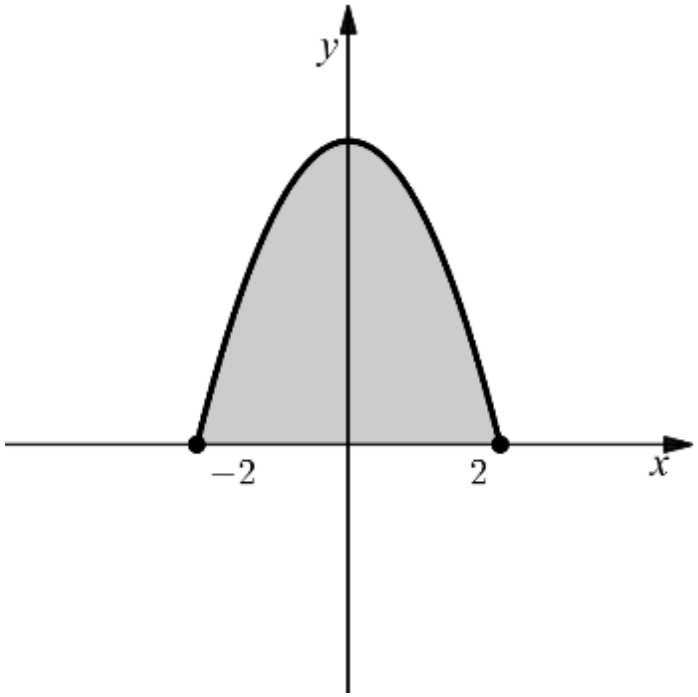
2 d)

$$\pi \int_1^4 y^2 \, \mathrm{d}x = \pi \int_1^4 (2 + \sqrt{x})^2 \, \mathrm{d}x = \pi \int_1^4 (4 + 4\sqrt{x} + x) \, \mathrm{d}x$$
$$= \pi \left[4x + \frac{8}{3} x^{\frac{3}{2}} + \frac{1}{2} x^2 \right]_1^4$$
$$= \pi \left[\left(4 \times 4 + \frac{8}{3} \times 4^{\frac{3}{2}} + \frac{1}{2} \times 4^2 \right) - \left(4 + \frac{8}{3} + \frac{1}{2} \right) \right]$$
$$= \frac{229}{6} \pi \text{ units}^3$$

3)

$$\pi \int_1^2 y^2 \, \mathrm{d}x = \pi \int_1^2 (x^3 + 2)^2 \, \mathrm{d}x = \pi \int_1^2 (x^6 + 4x^3 + 4) \, \mathrm{d}x$$
$$= \pi \left[\frac{1}{7} x^7 + x^4 + 4x \right]_1^2$$
$$= \pi \left[\left(\frac{2^7}{7} + 2^4 + 8 \right) - \left(\frac{1^7}{7} + 1^4 + 4 \right) \right]$$
$$= \frac{260}{7} \pi$$
$$= 116.7 \text{ cm}^3$$

4 a)



4 b)

$$\pi \int_{-2}^{2} y^2 \, \mathrm{d}x = \pi \int_{-2}^{2} (4 - x^2)^2 \, \mathrm{d}x = \pi \int_{-2}^{2} (16 - 8x^2 + x^4) \, \mathrm{d}x$$
$$= \pi \left[16x - \frac{8}{3}x^3 + \frac{1}{5}x^5 \right]_{-2}^{2}$$
$$= \pi \left[\left(32 - \frac{8}{3} \times 2^3 + \frac{1}{5} \times 2^5 \right) - \left(-32 - \frac{8}{3} \times (-2)^3 + \frac{1}{5} \times (-2)^5 \right) \right]$$
$$= \frac{512}{15}\pi \text{ units}^3$$

5)

$$\pi \int_{0}^{h} r^2 \, \mathrm{d}x$$
$$= \pi \left[r^2 x \right]_{0}^{h}$$
$$= \pi \left[(r^2 h) - (0) \right]$$
$$= \pi r^2 h \text{ as req.}$$

6 a)

$y = kx$ and $x = h$ intersect when $y = r$

$$\therefore k = \frac{r}{h}$$

6 b)

$$\pi \int_0^h y^2 \, \mathrm{d}x = \pi \int_0^h \left(\frac{r}{h}x\right)^2 \, \mathrm{d}x = \pi \int_0^h \frac{r^2}{h^2}x^2 \, \mathrm{d}x$$
$$= \pi \left[\frac{r^2}{3h^2}x^3\right]_0^h$$
$$= \pi \left[\left(\frac{r^2}{3h^2}h^3\right) - (0)\right]$$
$$= \frac{1}{3}\pi r^2 h \text{ as req.}$$

7 a)

$$\pi \int_1^3 y^2 \, \mathrm{d}x = \pi \int_1^3 (\sqrt{x-1})^2 \, \mathrm{d}x = \pi \int_1^3 (x-1) \, \mathrm{d}x$$
$$= \pi \left[\frac{1}{2}x^2 - x\right]_1^3$$
$$= \pi \left[\left(\frac{9}{2} - 3\right) - \left(\frac{1}{2} - 1\right)\right]$$
$$= 2\pi \text{ units}^3$$

7 b)

$$\pi \int_{-1}^8 y^2 \, \mathrm{d}x = \pi \int_{-1}^8 (2 + x^{\frac{1}{3}})^2 \, \mathrm{d}x = \pi \int_{-1}^8 (4 + 4x^{\frac{1}{3}} + x^{\frac{2}{3}}) \, \mathrm{d}x$$
$$= \pi \left[4x + 3x^{\frac{4}{3}} + \frac{3}{5}x^{\frac{5}{3}}\right]_{-1}^8$$
$$= \pi \left[\left(32 + 48 + \frac{96}{5}\right) - \left(-4 + 3 - \frac{3}{5}\right)\right]$$
$$= \frac{504}{5}\pi \text{ units}^3$$

7 c)

$$-2x^2 + 6x = 2x(3 - x)$$

The curve meets the axis when $x = 0$ and $x = 3$

$$\pi \int_0^3 y^2 \, \mathrm{d}x = \pi \int_0^3 (-2x^2 + 6x)^2 \, \mathrm{d}x = \pi \int_0^3 (4x^4 - 24x^3 + 36x^2) \, \mathrm{d}x$$
$$= \pi \left[\frac{4}{5}x^5 - 6x^4 + 12x^3\right]_0^3$$
$$= \pi \left[\left(\frac{972}{5} - 486 + 324\right) - (0)\right]$$
$$= \frac{162}{5}\pi \text{ units}^3$$

8)

$$x^3 - 4x^2 = x^2(x-4)$$

The curve meets the x-axis when $x = 0$ and $x = 4$

$$\begin{aligned}
&\pi \int_0^4 y^2 \,\mathrm{d}x = \pi \int_0^4 (x^3 - 4x^2)^2 \,\mathrm{d}x = \pi \int_0^4 (x^6 - 8x^5 + 16x^4) \,\mathrm{d}x \\
&= \pi \left[\frac{1}{7}x^7 - \frac{4}{3}x^6 + \frac{16}{5}x^5 \right]_0^4 \\
&= \pi \left[\left(\frac{16384}{7} - \frac{16384}{3} + \frac{16384}{5} \right) - (0) \right] \\
&= \frac{16384}{105}\pi \\
&= 490.2 \text{ units}^3
\end{aligned}$$

9 a)

$y = 1 + \frac{1}{1} = 2$ and $y = 1.5 + \frac{1}{1.5} = \frac{13}{6}$

So radii are 2 and $\frac{13}{6}$

Volume is $0.5 \times \pi \times 2^2 + 0.5 \times \pi \times \left(\frac{13}{6}\right)^2 = 13.7 \text{ units}^3$ (3 SF)

9 b)

He could increase the number of cylinders he is using.

9 c)

$$\begin{aligned}
&\pi \int_{0.5}^{1.5} y^2 \,\mathrm{d}x = \pi \int_{0.5}^{1.5} \left(x + \frac{1}{x} \right)^2 \,\mathrm{d}x = \pi \int_{0.5}^{1.5} (x^2 + 2 + x^{-2}) \,\mathrm{d}x \\
&= \pi \left[\frac{1}{3}x^3 + 2x - x^{-1} \right]_{0.5}^{1.5} \\
&= \pi \left[\left(\frac{1.5^3}{3} + 3 - 1.5^{-1} \right) - \left(\frac{0.5^3}{3} + 1 - 0.5^{-1} \right) \right] \\
&= \frac{53}{12}\pi \text{ units}^3
\end{aligned}$$

10 a)

$$\begin{aligned}
&\pi \int_0^r y^2 \,\mathrm{d}x = \pi \int_0^r (r^2 - x^2) \,\mathrm{d}x \\
&= \pi \left[r^2 x - \frac{1}{3}x^3 \right]_0^r \\
&= \pi \left[\left(r^3 - \frac{1}{3}r^3 \right) - (0) \right] \\
&= \frac{2}{3}\pi r^3 \text{ units}^3
\end{aligned}$$

10 b)

$$2 \times \frac{2}{3}\pi r^3 = \frac{4}{3}\pi r^3 \text{ as req.}$$

11 a)

When $y = 0$:

$$-6x = 9 - x^3$$
$$x^3 - 6x - 9 = 0$$

Let $\mathrm{f}(x) = x^3 - 6x - 9$

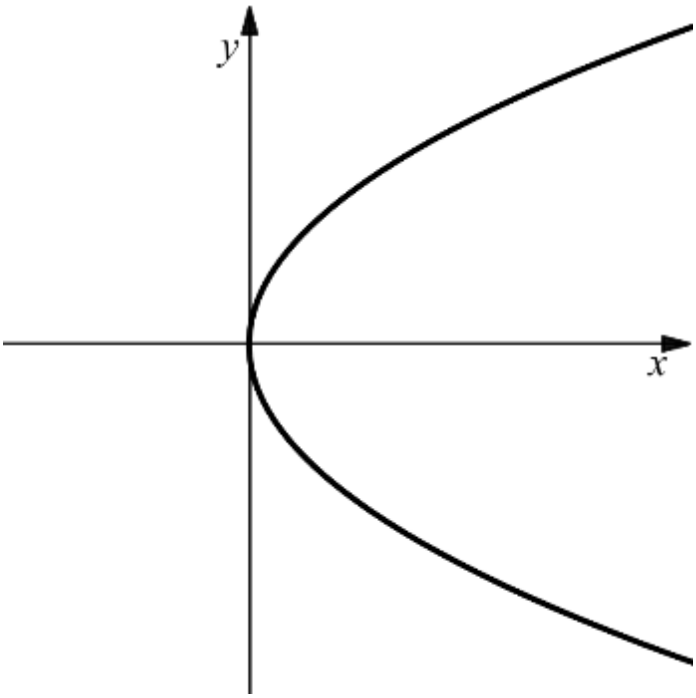
$\mathrm{f}(3) = 27 - 18 - 9 = 0$ so $(x - 3)$ is a factor so $x = 3$ is a solution.

The curve meets the x-axis when $x = 3$

11 b)

$$\pi \int_{-3}^{3} y^2 \, \mathrm{d}x = \pi \int_{-3}^{3} \left(\frac{9}{2} - \frac{1}{2}x^3 + 3x \right) \mathrm{d}x$$
$$= \pi \left[\frac{9}{2}x - \frac{1}{8}x^4 + \frac{3}{2}x^2 \right]_{-3}^{3}$$
$$= \pi \left[\left(\frac{27}{2} - \frac{81}{8} + \frac{27}{2} \right) - \left(-\frac{27}{2} - \frac{81}{8} + \frac{27}{2} \right) \right]$$
$$= 27\pi \text{ units}^3$$

12 a)



12 b)

$$\begin{aligned}\pi\int_0^d y^2\,\mathrm{d}x &= \pi\int_0^d 3x\,\mathrm{d}x\\ &= \pi\left[\frac{3}{2}x^2\right]_0^d\\ &= \pi\left[\left(\frac{3}{2}d^2\right)-0\right]\\ &= \frac{3}{2}\pi d^2\end{aligned}$$

$500 = \frac{3}{2}\pi d^2$ so $d = \sqrt{\frac{1000}{3\pi}} = 10.3$ cm

12 c)

That the material the bowl is has no thickness

13 a)

$$\begin{aligned}180 = \pi\int_0^6 y^2\,\mathrm{d}x &= \pi\int_0^6 kx\,\mathrm{d}x\\ &= \pi\left[\frac{k}{2}x^2\right]_0^6\\ &= \pi[(18k)-0]\\ &= 13\pi k\end{aligned}$$

$180 = 36\pi k$ so $k = \frac{10}{\pi}$

So the model is $y^2 = \frac{10}{\pi}x$

The radius occurs when $x = 6$ so $r = \sqrt{\frac{10}{\pi}\times 6} = 4.4$ cm

13 b)

Reduce the value of k and increase the height of the glass using the formula

14)

$$\begin{aligned}\pi\int_1^{\sqrt{3}} y^2\,\mathrm{d}x = \pi\int_1^{\sqrt{3}}(-x^2+3)^2\,\mathrm{d}x &= \pi\int_1^{\sqrt{3}} x^4-6x^2+9\,\mathrm{d}x\\ &= \pi\left[\frac{1}{5}x^5-2x^3+9x\right]_1^{\sqrt{3}}\\ &= \pi\left[\left(\frac{1}{5}\times\sqrt{3}^5-2\times\sqrt{3}^3+9\sqrt{3}\right)-\left(\frac{1}{5}-2+9\right)\right]\\ &= \pi\left[\left(\frac{9}{5}\sqrt{3}-6\sqrt{3}+9\sqrt{3}\right)-\left(\frac{1}{5}-2+9\right)\right]\\ &= \left(\frac{24}{5}\sqrt{3}-\frac{36}{5}\right)\pi\end{aligned}$$

Volume of cone: $\frac{1}{3}\pi\times 2^2\times 1 = \frac{4}{3}\pi$

Total volume: $\frac{4}{3}\pi + \left(\frac{24}{5}\sqrt{3}-\frac{36}{5}\right)\pi = \left(\frac{24}{5}\sqrt{3}-\frac{88}{15}\right)\pi$ units^3

15 a)

$y = 8 - 3x$ so $A\left(\frac{8}{3}, 0\right)$

$$\frac{1}{4}x^3 = 8 - 3x \Rightarrow x^3 = 32 - 12x \Rightarrow x^3 + 12x - 32 = 0$$

Let $\mathrm{f}(x) = x^3 + 12x - 32$

$\mathrm{f}(2) = 8 + 24 - 32 = 0$ so $(x - 2)$ is a factor so $x = 2$ is a solution so $B(2, 2)$

$A\left(\frac{8}{3}, 0\right)$ and $B(2, 2)$

15 b)

$$\begin{aligned}
&\pi \int_0^2 y^2 \, \mathrm{d}x = \pi \int_0^2 \frac{1}{16}x^6 \, \mathrm{d}x \\
&= \pi \left[\frac{1}{112}x^7\right]_0^2 \\
&= \pi \left[\frac{8}{7} - 0\right] \\
&= \frac{8}{7}\pi
\end{aligned}$$

Volume of cone: $\frac{1}{3}\pi \times 2^2 \times \left(\frac{8}{3} - 2\right) = \frac{8}{9}\pi$

Total volume: $\frac{8}{7}\pi + \frac{8}{9}\pi = \frac{128}{63}\pi$ units^3

16 a)

$$\begin{aligned}
2x + 0.5x^2 &= 6 \\
0.5x^2 + 2x - 6 &= 0
\end{aligned}$$

Gives $x = 2$ or $x = -6$ so $x = 2$

$(2, 2)$

16 b)

$$\begin{aligned}
&\pi \int_2^3 y^2 \, \mathrm{d}x = \pi \int_2^3 \frac{1}{4}x^4 \, \mathrm{d}x \\
&= \pi \left[\frac{1}{20}x^5\right]_2^3 \\
&= \pi \left[\frac{243}{20} - \frac{8}{5}\right] \\
&= \frac{211}{20}\pi
\end{aligned}$$

Volume of cone: $\frac{1}{3}\pi \times 2^2 \times (3 - 2) = \frac{4}{3}\pi$

Total volume: $\frac{211}{20}\pi - \frac{4}{3}\pi = \frac{553}{60}\pi$ units^3

17 a)

$$\begin{aligned}
&\frac{1}{4} \times \pi \int_1^4 y^2 \, \mathrm{d}x = \frac{\pi}{4} \int_1^4 (\frac{1}{4}x^2 + x + 1) \, \mathrm{d}x \\
&= \frac{\pi}{4} \left[\frac{1}{12}x^3 + \frac{1}{2}x^2 + x \right]_1^4 \\
&= \frac{\pi}{4} \left[\left(\frac{16}{3} + 8 + 4 \right) - \left(\frac{1}{12} + \frac{1}{2} + 1 \right) \right] \\
&= \frac{63}{16}\pi \text{ units}^3
\end{aligned}$$

17 b)

$$\begin{aligned}
&\frac{1}{2} \times \pi \int_1^2 y^2 \, \mathrm{d}x = \frac{\pi}{2} \int_1^2 (x^4 - 2x^2 + 1) \, \mathrm{d}x \\
&= \frac{\pi}{2} \left[\frac{1}{5}x^5 - \frac{2}{3}x^3 + x \right]_1^2 \\
&= \frac{\pi}{2} \left[\left(\frac{32}{5} - \frac{16}{3} + 2 \right) - \left(\frac{1}{5} - \frac{2}{3} + 1 \right) \right] \\
&= \frac{19}{15}\pi \text{ units}^3
\end{aligned}$$

17 c)

$$\begin{aligned}
&\frac{1}{4} \times \pi \int_0^1 y^2 \, \mathrm{d}x = \frac{\pi}{4} \int_0^1 (16 - 8\sqrt{x} + x) \, \mathrm{d}x \\
&= \frac{\pi}{4} \left[16x - \frac{16}{3}x^{\frac{3}{2}} + \frac{1}{2}x^2 \right]_0^3 \\
&= \frac{\pi}{4} \left[\left(16 - \frac{16}{3} + \frac{1}{2} \right) - (0) \right] \\
&= \frac{67}{24}\pi \text{ units}^3
\end{aligned}$$

18)

$$\begin{aligned}
&\pi \int_{-4}^4 y^2 \, \mathrm{d}x = \pi \int_{-4}^4 (16 - x^2) \, \mathrm{d}x \\
&= \pi \left[16x - \frac{1}{3}x^3 \right]_{-4}^4 \\
&= \pi \left[\left(64 - \frac{64}{3} \right) - \left(-64 + \frac{64}{3} \right) \right] \\
&= \frac{256}{3}\pi
\end{aligned}$$

and

$$\pi \int_{-2}^2 y^2 \, \mathrm{d}x = \pi \int_{-2}^2 2^2 \, \mathrm{d}x$$

$$= \pi \left[4x\right]_{-2}^{2}$$
$$= \pi \left[8 - (-8)\right]$$
$$= 16\pi$$

Total volume: $\frac{256}{3}\pi - 16\pi = \frac{208}{3}\pi$ units^3

19 a)

$$2x = -x^2 + 4x \Rightarrow x^2 - 4x + 2x = 0 \Rightarrow x^2 - 2x = 0 \Rightarrow x(x-2) = 0$$

So $x = 2$

$A(2, 4)$

19 b)

$B(2, -4)$

19 c)

Volume of cone: $\frac{1}{3}\pi \times 4^2 \times 2 = \frac{32}{3}\pi$

and

$$\pi \int_2^4 y^2 \, \mathrm{d}x = \pi \int_2^4 (-x^2 + 4x)^2 \, \mathrm{d}x = \pi \int_2^4 (x^4 - 8x^3 + 16x^2) \, \mathrm{d}x$$
$$= \pi \left[\frac{1}{5}x^5 - 2x^4 + \frac{16}{3}x^3\right]_2^4$$
$$= \pi \left[\left(\frac{1024}{5} - 512 + \frac{1024}{3}\right) - \left(\frac{32}{5} - 32 + \frac{128}{3}\right)\right]$$
$$= \frac{256}{15}\pi$$

Total volume: $\frac{32}{3}\pi + \frac{256}{15}\pi = \frac{416}{15}\pi$ units^3

20)

$$\pi \int_{-3}^{3} y^2 \, \mathrm{d}x = \pi \int_{-3}^{3} (9 - x^2) \, \mathrm{d}x$$
$$= \pi \left[9x - \frac{1}{3}x^3\right]_{-3}^{3}$$
$$= \pi \left[(27 - 9) - (-27 + 9)\right]$$
$$= 36\pi$$

and

$$\pi \int_{-2}^{2} y^2 \, \mathrm{d}x = \pi \int_{-2}^{2} (4 - x^2) \, \mathrm{d}x$$
$$= \pi \left[4x - \frac{1}{3}x^3\right]_{-2}^{2}$$
$$= \pi \left[\left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right)\right]$$
$$= \frac{32}{3}\pi$$

Total volume: $36\pi - \frac{32}{3}\pi = \frac{76}{3}\pi$ units^3

21 a)

$$0 = -\frac{1}{8}x^2 + 2 \Rightarrow x = 4$$

$A(4, 0)$

21 b)

$$\begin{aligned}
&\pi \int_0^4 y^2 \, \mathrm{d}x = \pi \int_0^4 (16 - x^2) \, \mathrm{d}x \\
&= \pi \left[16x - \frac{1}{3}x^3 \right]_0^4 \\
&= \pi \left[\left(64 - \frac{64}{3} \right) - (0) \right] \\
&= \frac{128}{3}\pi
\end{aligned}$$

and

$$\begin{aligned}
&\pi \int_0^4 y^2 \, \mathrm{d}x = \pi \int_0^4 (-\frac{1}{8}x^2 + 2)^2 \, \mathrm{d}x = \pi \int_0^4 (\frac{1}{64}x^4 - \frac{1}{2}x^2 + 4) \, \mathrm{d}x \\
&= \pi \left[\frac{1}{320}x^5 - \frac{1}{6}x^3 + 4x \right]_0^4 \\
&= \pi \left[\left(\frac{1}{320} \times 4^5 - \frac{1}{6} \times 4^3 + 16 \right) - (0) \right] \\
&= \frac{128}{15}\pi
\end{aligned}$$

Total volume: $\frac{128}{3}\pi - \frac{128}{15}\pi = \frac{512}{15}\pi$ units^3

22 a)

$$\begin{aligned}
&\pi \int_1^5 x^2 \, \mathrm{d}y = \pi \int_1^5 (y^2 + 4y + 4) \, \mathrm{d}y \\
&= \pi \left[\frac{1}{3}y^3 + 2y^2 + 4y \right]_1^5 \\
&= \pi \left[\left(\frac{125}{3} + 50 + 20 \right) - \left(\frac{1}{3} + 2 + 4 \right) \right] \\
&= \frac{316}{3}\pi \text{ units}^3
\end{aligned}$$

22 b)

$$\begin{aligned}
&\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 y^6 \, \mathrm{d}y \\
&= \pi \left[\frac{1}{7}y^7 \right]_0^1 \\
&= \pi \left[\frac{1}{7} - 0 \right] \\
&= \frac{1}{7}\pi \text{ units}^3
\end{aligned}$$

22 c)

$$\begin{aligned}
&\pi \int_0^4 x^2 \,\mathrm{d}y = \pi \int_0^4 y \,\mathrm{d}y \\
&= \pi \left[\frac{1}{2}y^2\right]_0^4 \\
&= \pi \left[8 - 0\right] \\
&= 8\pi \text{ units}^3
\end{aligned}$$

23)

$$\begin{aligned}
&\pi \int_4^{10} x^2 \,\mathrm{d}y = \pi \int_4^{10} (y + 6) \,\mathrm{d}y \\
&= \pi \left[\frac{1}{2}y^2 + 6y\right]_4^{10} \\
&= \pi \left[\left(\frac{1}{2} \times 10^2 + 60\right) - \left(\frac{1}{2} \times 4^2 + 24\right)\right] \\
&= 78\pi \text{ as req.}
\end{aligned}$$

24)

$$\begin{aligned}
&\pi \int_0^4 x^2 \,\mathrm{d}y = \pi \int_0^4 (y - y^{\frac{1}{2}} + 1) \,\mathrm{d}y \\
&= \pi \left[\frac{1}{2}y^2 - \frac{2}{3}y^{\frac{3}{2}} + y\right]_0^4 \\
&= \pi \left[\left(8 - \frac{16}{3} + 4\right) - (0)\right] \\
&= \frac{20}{3}\pi \text{ units}^3
\end{aligned}$$

25)

$$\begin{aligned}
&\pi \int_1^4 x^2 \,\mathrm{d}y = \pi \int_1^4 (y^2 - 4y + 5)^2 \,\mathrm{d}y = \pi \int_1^4 (y^4 - 8y^3 + 26y^2 - 40y + 25) \,\mathrm{d}y \\
&= \pi \left[\frac{1}{5}y^5 - 2y^4 + \frac{26}{3}y^3 - 20y^2 + 25y\right]_1^4 \\
&= \pi \left[\left(\frac{1024}{5} - 512 + \frac{1664}{3} - 320 + 100\right) - \left(\frac{1}{5} - 2 + \frac{26}{3} - 20 + 25\right)\right] \\
&= \frac{78}{5}\pi \text{ units}^3
\end{aligned}$$

26)

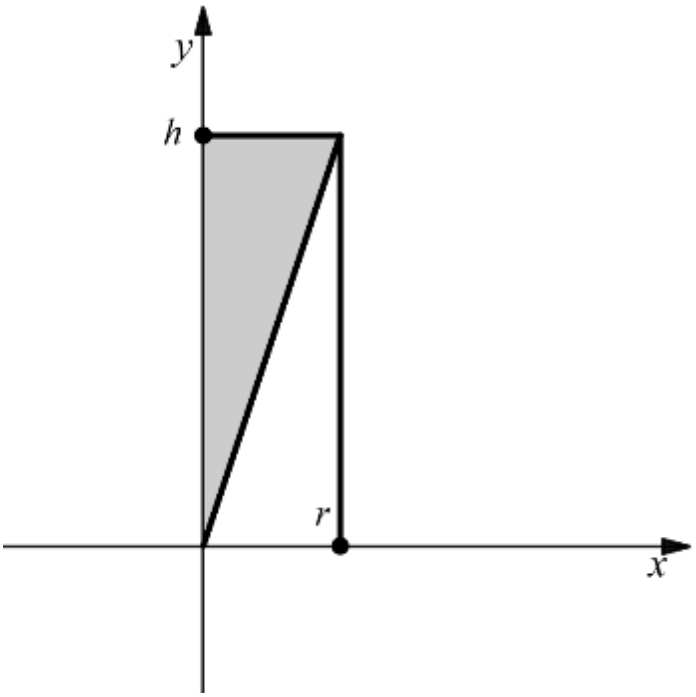
$$\pi \int_0^1 x^2 \,\mathrm{d}y = \pi \int_0^1 (3 - y^3 - 2y) \,\mathrm{d}y$$

$$= \pi \left[3y - \frac{1}{4}y^4 - y^2 \right]_0^1$$
$$= \pi \left[\left(3 - \frac{1}{4} - 1 \right) - (0) \right]$$
$$= \frac{7}{4}\pi \text{ units}^3$$

27)

$$\pi \int_0^h x^2 \, \mathrm{d}y = \pi \int_0^h r^2 \, \mathrm{d}y$$
$$= \pi \left[r^2 y \right]_0^h$$
$$= \pi \left[(r^2 h) - (0) \right]$$
$$= \pi r^2 h \text{ units}^3$$

28)



$y = kx$ and $y = h$ intersect when $x = r$

$$\therefore k = \frac{h}{r}$$

$$\pi \int_0^h x^2 \, \mathrm{d}y = \pi \int_0^h \left(\frac{y}{k} \right)^2 \mathrm{d}y = \pi \int_0^h \frac{r^2}{h^2} y^2 \, \mathrm{d}y$$
$$= \pi \left[\frac{r^2}{3h^2} y^3 \right]_0^h$$
$$= \pi \left[\left(\frac{r^2}{3h^2} h^3 \right) - (0) \right]$$
$$= \frac{1}{3}\pi r^2 h \text{ as req.}$$

29 a)

$A(-1, 0)$, $B(1, 0)$ and $C(0, 1)$

29 b)

$$\begin{aligned}
&\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 1 - y \, \mathrm{d}y \\
&= \pi \left[y - \frac{1}{2} y^2 \right]_0^1 \\
&= \pi \left[\left(1 - \frac{1}{2} \right) - (0) \right] \\
&= \frac{1}{2} \pi \approx 1.57 \text{ m}^3
\end{aligned}$$

29 c)

$$1600 \times \frac{1}{2} \pi = 2510 \text{ kg}$$

29 d)

This will only be true if the sand is perfectly symmetrical and follows the curve exactly. There may be inconsistencies in the sand, such as varying compactness so that the density is not the same throughout.

30 a)

$$V = \pi \int_0^d x^2 \, \mathrm{d}y$$

$$\begin{aligned}
&\pi \int_0^d \left(\frac{y}{0.4} \right) \mathrm{d}y \\
&= \pi \left[\frac{y^2}{0.8} \right]_0^d \\
&= \pi \left[\left(\frac{d^2}{0.8} \right) - 0 \right] \\
&= \frac{\pi}{0.8} d^2 \\
&\approx 3.93 d^2 \text{ as req.}
\end{aligned}$$

30 b)

$3.93 \times 6^2 = 141 \text{ cm}^3$ (3 SF)

30 c)

$360 = 3.93 d^2$ gives $d \approx 9.57$ cm

31 a)

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 4^2 \times 6 = 32\pi \approx 101 \text{ cm}^3$

31 b)

$$\frac{4}{6}r = \frac{2}{3}h \text{ so } V = \frac{1}{3}\pi\left(\frac{2}{3}h\right)^2 h$$
$$\frac{32\pi}{2} = \frac{1}{2}\pi\left(\frac{2}{3}h\right)^2 h$$
$$48 = \frac{4}{9}h^3$$
$$h^3 = 108$$
$$h = 4.76 \text{ cm}$$

31 c)

Gradient: $\frac{6}{4} = \frac{3}{2}$ and goes through the y-axis at $(0, 8)$

$$y = \frac{3}{2}x + 8 \Rightarrow y - 8 = \frac{3}{2}x$$
$$\therefore x = \frac{2y - 16}{3}$$

31 d)

$$\pi\int_8^{14} x^2\,\mathrm{d}y = \pi\int_8^{14}\left(\frac{2}{3}(y-8)\right)^2\,\mathrm{d}y = \pi\int_8^{14}(\frac{4}{9}y^2 - \frac{64}{9}y + \frac{256}{9})\,\mathrm{d}y$$
$$= \pi\left[\frac{4}{27}y^3 - \frac{32}{9}y^2 + \frac{256}{9}y\right]_8^{14}$$
$$= \pi\left[\left(\frac{4}{27}\times 14^3 - \frac{32}{9}\times 14^2 + \frac{256}{9}\times 14\right) - \left(\frac{4}{27}\times 8^3 - \frac{32}{9}\times 8^2 + \frac{256}{9}\times 8\right)\right]$$
$$= 32\pi \approx 101 \text{ cm}^3$$

32)

Volume of cone: $\frac{1}{3}\pi \times 2^2 \times (8-6) = \frac{8\pi}{3}$

$$\pi\int_0^6 x^2\,\mathrm{d}y = \pi\int_0^6 \frac{y}{1.5}\,\mathrm{d}y$$
$$= \pi\left[\frac{1}{3}y^2\right]_0^6$$
$$= \pi\left[\left(\frac{6^2}{3}\right) - (0)\right]$$
$$= 12\pi$$

Total volume is $\frac{44\pi}{3}$ units^3

33 a)

$$3.2 = k\sqrt{16} - 0.05 \times 16^2$$

Gives $k = 4$

33 b)

Cylinder: $\pi \times 3.2^2 \times 2 = \frac{512}{25}\pi$

$$\begin{aligned}
&\pi \int_0^{16} x^2 \, \mathrm{d}y = \pi \int_0^{16} (4\sqrt{y} - 0.05y^2)^2 \, \mathrm{d}y = \pi \int_0^{16} 16y - \frac{2}{5}y^{\frac{5}{2}} + \frac{1}{400}y^4 \, \mathrm{d}y \\
&= \pi \left[8y^2 - \frac{4}{35}y^{\frac{7}{2}} + \frac{1}{2000}y^5 \right]_0^{16} \\
&= \pi \left[\left(2048 - \frac{65536}{35} + \frac{65536}{125} \right) - (0) \right] \\
&= \frac{612352}{875}\pi
\end{aligned}$$

Volume: $\frac{512}{25}\pi + \frac{612352}{875}\pi = 2260 \text{ cm}^3$ (3 SF)

33 c)

i)

The curve may not represent the shape of the vase accurately, for example, the curve does not flatten out to give a surface for the vase to stand on.

ii)

The volume calculation does not allow for the thickness of the material that it is made out of.

34)

$$y = x^2 - 4x + 4 = (x - 2)^2$$

So that $\sqrt{y} = x - 2$

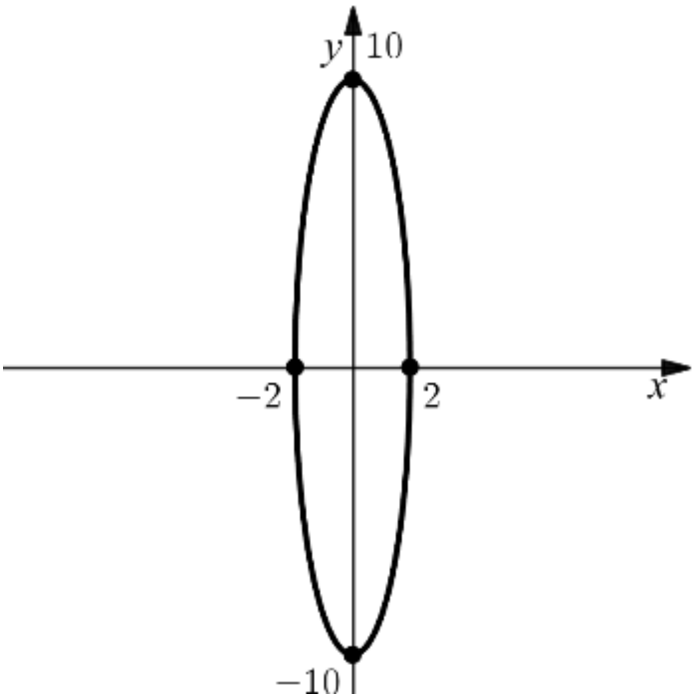
$\therefore x = 2 + \sqrt{y}$ and so $x^2 = 4 + 4\sqrt{y} + y$

$$\begin{aligned}
&\pi \int_0^3 x^2 \, \mathrm{d}y = \pi \int_0^3 4 + 4\sqrt{y} + y \, \mathrm{d}y \\
&= \pi \left[4y + \frac{8}{3}y^{\frac{3}{2}} + \frac{1}{2}y^2 \right]_0^3 \\
&= \pi \left[\left(12 + 8\sqrt{3} + \frac{9}{2} \right) - (0) \right] \\
&= \left(\frac{33}{2} + 8\sqrt{3} \right) \pi
\end{aligned}$$

35 a)

When $x = 0$, $\frac{y^2}{100} = 1$ so $y = \pm 10$

When $y = 0$, $\frac{x^2}{4} = 1$ so $y = \pm 2$



35 b)

$\frac{y^2}{100} = 1 - \frac{x^2}{4}$ so $y^2 = 100 - 25x^2$

$$\begin{aligned}
\pi \int_{-2}^{2} y^2 \, \mathrm{d}x &= \pi \int_{-2}^{2} 100 - 25x^2 \, \mathrm{d}x \\
&= \pi \left[100x - \frac{25}{3}x^3 \right]_{-2}^{2} \\
&= \pi \left[\left(200 - \frac{200}{3} \right) - \left(-200 + \frac{200}{3} \right) \right] \\
&= \frac{800}{3}\pi \text{ units}^3
\end{aligned}$$

35 c)

$\frac{x^2}{4} = 1 - \frac{y^2}{100}$ so $x^2 = 4 - \frac{1}{25}y^2$

$$\begin{aligned}
\pi \int_{-10}^{10} x^2 \, \mathrm{d}y &= \pi \int_{-10}^{10} 4 - \frac{1}{25}y^2 \, \mathrm{d}y \\
&= \pi \left[4y - \frac{1}{75}y^3 \right]_{-10}^{10} \\
&= \pi \left[\left(40 - \frac{40}{3} \right) - \left(-40 + \frac{40}{3} \right) \right] \\
&= \frac{160}{3}\pi \text{ units}^3
\end{aligned}$$

36 a)

$169y^2 = 676 - 25x^2$ so $y^2 = 4 - \frac{25}{169}x^2$

$$\begin{aligned}
&\pi \int_{-5.2}^{5.2} y^2 \, \mathrm{d}x = \pi \int_{-5.2}^{5.2} 4 - \frac{25}{169}x^2 \, \mathrm{d}x \\
&= \pi \left[4x - \frac{25}{507}x^3 \right]_{-5.2}^{5.2} \\
&= \pi \left[\left(\frac{104}{5} - \frac{104}{15} \right) - \left(-\frac{104}{5} + \frac{104}{15} \right) \right] \\
&= \frac{416}{15}\pi \\
&= 87.1 \text{ mm}^3 \text{ (3 SF)}
\end{aligned}$$

36 b)

$25x^2 = 676 - 169y^2$ so $x^2 = \frac{676}{25} - \frac{169}{25}y^2$

$$\begin{aligned}
&\pi \int_{-2}^{2} x^2 \, \mathrm{d}y = \pi \int_{-2}^{2} \frac{676}{25} - \frac{169}{25}y^2 \, \mathrm{d}y \\
&= \pi \left[\frac{676}{25}y - \frac{169}{75}y^3 \right]_{-2}^{2} \\
&= \pi \left[\left(\frac{1352}{25} - \frac{1352}{75} \right) - \left(-\frac{1352}{25} + \frac{1352}{75} \right) \right] \\
&= \frac{5408}{75}\pi \\
&= 227 \text{ mm}^3 \text{ (3 SF)}
\end{aligned}$$

36 c)

Rotating about the y-axis (part b) gives a shape that better matches the actual geometry of the chocolate — round and flattened, with circular top view and shorter height.